Atomic Display Base

SKU:A115

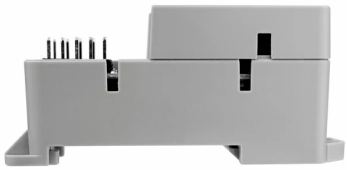








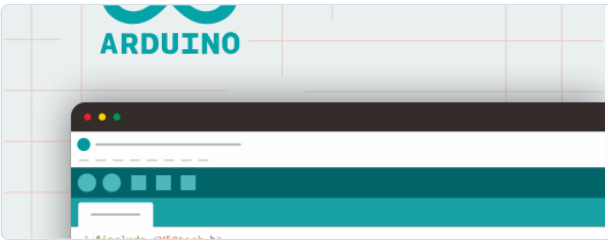




Description

Atomic Display Base is a high-definition monitor driver unit that uses an FPGA to simulate traditional SPI TFT-LCD data output, supporting a maximum 720P (1280 x 720) image output. It incorporates a dedicated RGB-to-HD multimedia signal conversion chip for wide compatibility in display output. Users can pair it with Atom series hosts to meet different memory and application requirements. The Atomic Display Base can replace the PC control scheme of traditional display panels, providing a more cost-effective solution. It is suitable for industrial control displays, smart home information screens, educational and conference presentations, and remote monitoring displays.

Tutorial



Arduino IDE

This tutorial will introduce how to program and control the Atomic Display Base device through Arduino IDE

Features

- Equipped with FPGA (Gowin GW1NR-9C)
- Built-in LT8618SX RGB-to-HD multimedia signal conversion chip (supports 24-bit color depth)
- SPI interface (FPGA) + I2C interface (LT8618SX)
- Maximum 720P (1280x720) image output
- Multiple output modes, optimized frame rate up to 12 ~ 16FPS
- Development Platforms: Arduino, ESP32-IDF (UIFlow not supported yet)

Includes

- 1 x Atomic Display Base

Applications

- Monitor input signal source
- High-definition data dashboard

Specifications

Specification	Parameter
FPGA	Gowin GW1NR-9C
LT8618SX	RGB-to-HD multimedia signal chip, supports 24-bit color
Max Image Output	720P (1280x720)
Output Frame Rate	1280x720 60Hz
Product Size	64.0 x 24.0 x 29.0mm
Package Size	136.0 x 92.0 x 30.0mm
Product Weight	15.2g
Gross Weight	17.3g

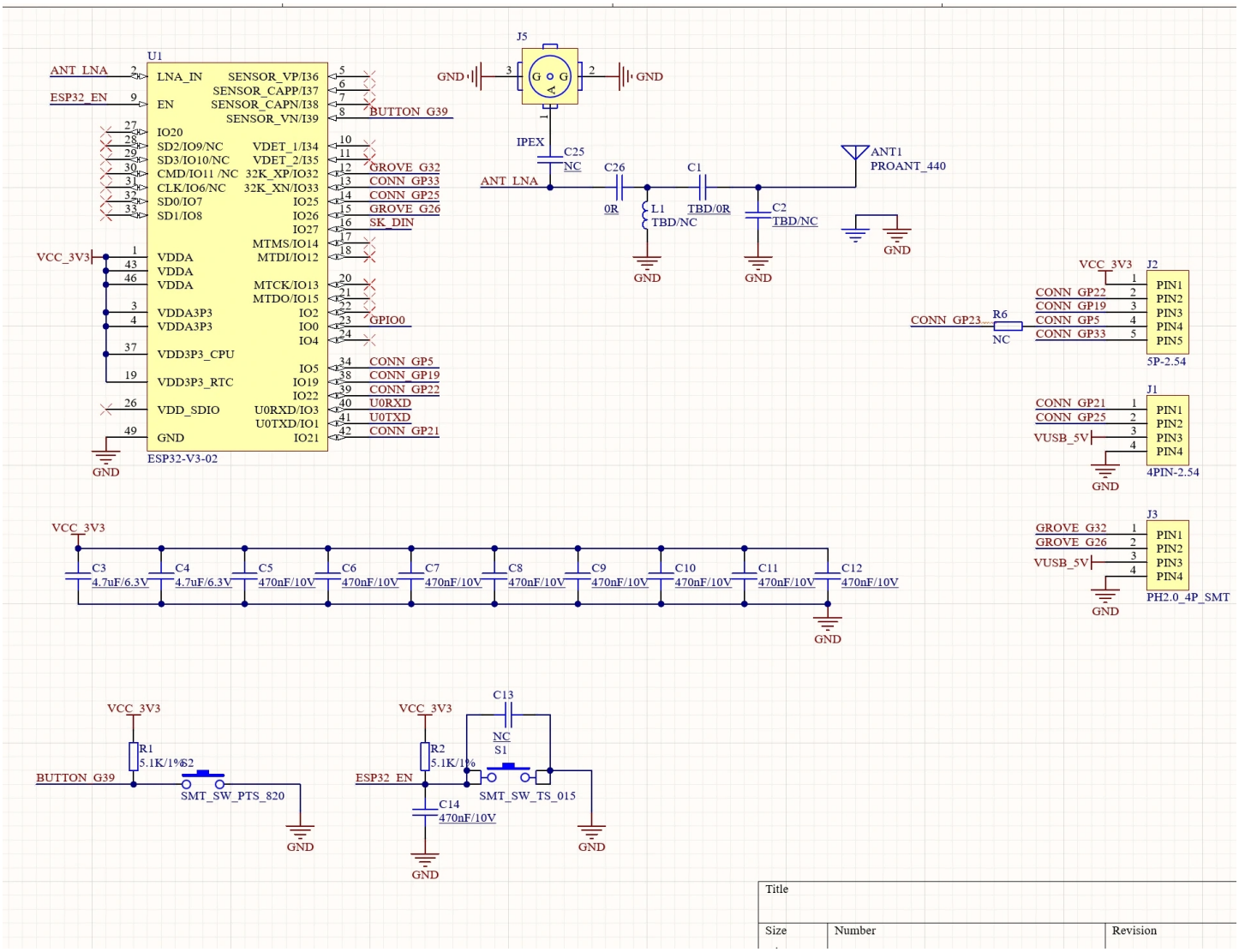
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Monitor Compatibility

Atomic Display Base must be used with monitors featuring adaptive resolution scaling functionality. On some monitors that do not support adaptive resolution, display anomalies may occur.

Schematics

- Atomic Display Base Schematics PDF



PinMap

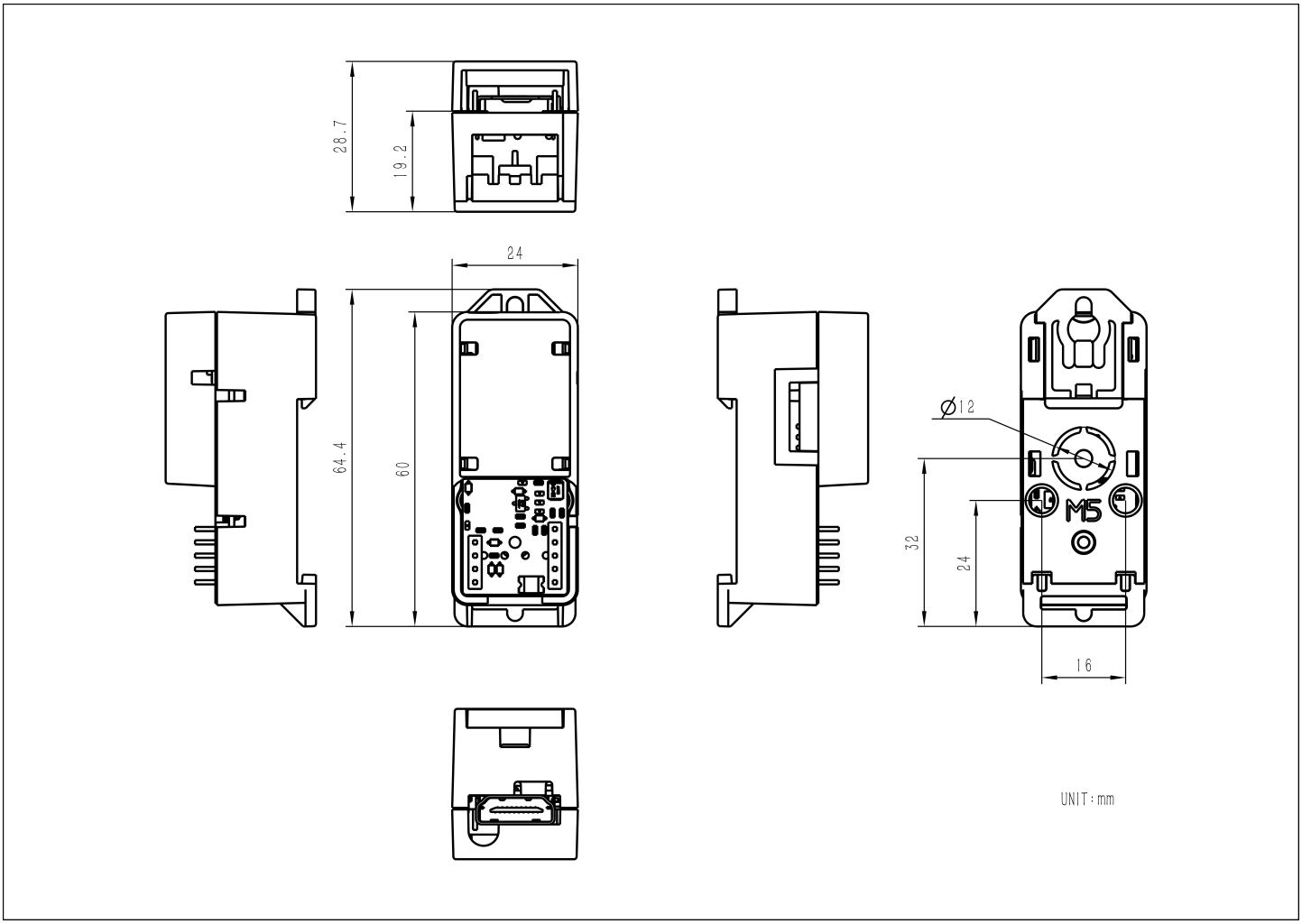
PIN	LEFT	RIGHT	PIN
		1	3V3
	2	3	SPI_CLK
	4	5	CS
5V	6	7	SPI_MISO
GND	8	9	SPI_MOSI

LT8618SX

Atomic Display Base	G25	G21
LT8618SX	LT_CSDA	LT_CSCL

Model Size

- Atomic Display Base Model Size PDF



Datasheets

- LT8618SX
- GW1NR-9C Data Sheet
- GW1NR-9C Design Manual

Softwares

Arduino

- M5GFX Library

UiFlow2

- Atomic Display Base UiFlow2 Docs

Video

- Atomic Display Base Product Demo

atomic display base video.mp4